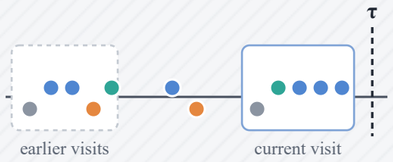
Input: patient history up to τ



τ = end of current visit

Lab results	353 types
Vital signs	6 types
ICD-10 codes	440 codes
Visits	4 types
Age, sex	2 prefix tokens

Each event = (key, value, time)

- Creatinine · 1.4 mg/dL · τ − 2 h
- NT-proBNP · 820 pg/mL · τ − 2 h
- Inpatient visit · – · τ − 3 d
- I10 · – · τ − 40 d

≤ 256 most recent events;
keeps ≤ 128 recent ICD/visit tokens
+ [AGE], [SEX]

Event embedding

Key embedding
$E_{key}(k)$

Type embedding
$E_{type}(c)$, 6 types

Value MLP
MLP(rank-Gauss(v))*

Time MLP
MLP(log(1 + Δt / 1 h))

Current-visit flag
$E_{cur}(0 / 1)$

+

$e_i \in \mathbb{R}^{96}$

** rank-Gauss fitted per key on training patients; learned vector if v is missing or masked ([MASK])*

SeqLab encoder

AGE | SEX | e_1 | e_2 | e_3 | … | e_L

3 × Transformer layer (pre-LN)

- Multi-head self-attention · 4 heads
- Feed-forward 96 → 384 → 96 · GELU
- Dropout 0.2 · residual

h_1 | h_2 | h_3 | … | h_L

Attention pooling
α = softmax(h·q / √d), q learned

$z = \text{LayerNorm}(\Sigma\, \alpha_i\, h_i) \in \mathbb{R}^{96}$
patient representation

$d = 96 \cdot L \le 256 + 2$

① Self-supervised pretraining (no diagnosis labels)

Masked-value head
$\text{Linear}(h_i) \to \hat{v}_i$
20% of values masked
MSE

Next-draw forecast head
$\hat{v} = v_{last} + \text{MLP}([h_{last}; z])$
first draw within 30 days
MSE · unseen keys: Linear(z)

Masked-ICD head
$\text{Linear}(h_i) \to$ ICD code
15% of ICD tokens masked
cross-entropy · 3 of 21 members

$\text{Loss} = \text{MSE}_{mask} + \text{MSE}_{forecast}\ (+\ \text{CE}_{ICD})$ · AdamW 3e-4, one-cycle, 6,000 steps

↓ pretrained encoder weights initialise ②

② Supervised fine-tuning for T2

T2 head
Dropout 0.2 → Linear(96 → 7) → sigmoid

- BCE over 7 codes · AdamW 2e-4 · whole encoder trained
- early stopping on validation macro AP
- multi-task members: 14 outputs (T2 + T3), T2 outputs used

I10 | **I50** | **I21** | **I48** | **I25** | **I20** | **I46**

σ(·) per code, multi-label: probabilities need not sum to 1

summarise

× 21 (seeds / variants)

Final T2 system

Summary features
451 columns: 60 labs × 5 history statistics, current-visit labs, codes, vitals, visit type, age, sex

HGB
5 bags per code

p_{HGB}

Logit blend
$p = \sigma(w \cdot \text{logit}\ p_{SeqLab} + (1 - w) \cdot \text{logit}\ p_{HGB})$
w ∈ {0, 0.25, 0.5, 0.75, 1}, chosen on validation

p_{SeqLab}

SeqLab ensemble · 21 networks

- 9 × single-task
- 3 × ICD-masked pretraining
- 9 × multi-task T2 + T3

→ mean probability

Output: probability that each code is newly recorded at the next visit

I10 Hypertension	**I50** Heart failure	**I21** Acute MI	**I48** AF / flutter	**I25** Chronic IHD	**I20** Angina	**I46** Cardiac arrest

Primary metric: macro AP over I10, I50, I21, I25, I20 (I48, I46 too rare)